

Homework 3

*Instructor: Henry Lam***Due Date:** Oct 3rd, 2025

Problem 1 Suppose that the loss L , due to the arrival of certain type of insurance claims, is distributed according to a Pareto distribution. Specifically, it has been observed that the distribution of L is satisfactorily modeled by the cumulative distribution function,

$$P(L \leq x) = \begin{cases} 1 - 1/x^3, & \text{if } x > 1, \\ 0, & \text{otherwise.} \end{cases}$$

Given access only to a procedure that generates $\text{Unif}(0,1)$, write an algorithm to generate L . Implement the algorithm on a computer to simulate 100 copies of L and plot their distribution.

Problem 2 We consider generating a random variable having density function

$$f(x) = \frac{e^x}{e-1}, 0 < x < 1.$$

- (a) Write the inverse transform method for generating this random variable.
- (b) Write the acceptance-rejection method, with a proposal density $\text{Unif}[0, 1]$.
- (c) Which method in part (a) or (b) do you prefer? Explain your answer.
- (d) Implement the algorithms in both parts (a) and (b) on a computer. Using each algorithm, simulate 100 copies of L and plot their distribution.

Problem 3 A Gamma distribution with parameters $(n, 1)$ (denoted by $\text{Gamma}(n, 1)$) has density function

$$f(x) = \begin{cases} e^{-x} \frac{x^{n-1}}{(n-1)!}, & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

- (a) Derive the acceptance-rejection method to generate a random variable X with distribution $\text{Gamma}(n, 1)$, by using an exponential density function with parameter $1/2$, i.e.,

$$g(x) = \frac{1}{2}e^{-x/2}, \quad x \geq 0.$$

Assume you have access only to a procedure that can generate $\text{Unif}(0,1)$.

- (b) Do you think using an exponential density with parameter 1 as the proposal would work in part (a)? Explain your answer.

Problem 4 The random variable X has a *semi-circular* distribution, with the probability density function (p.d.f) $f(x)$ given by

$$f(x) = \begin{cases} \frac{2}{\pi} \sqrt{1-x^2} & \text{if } -1 \leq x \leq 1; \\ 0 & \text{otherwise.} \end{cases}$$

A plot of the p.d.f is given below.

- (a) Derive the acceptance-rejection method to generate the random variable X , by using $\text{Unif}(-1,1)$ as the proposal. Assume you have access only to a procedure that can generate $\text{Unif}(0,1)$.
- (b) What is the expected runtime of the method, in terms of the number of acceptance-rejection trials needed to get one output?

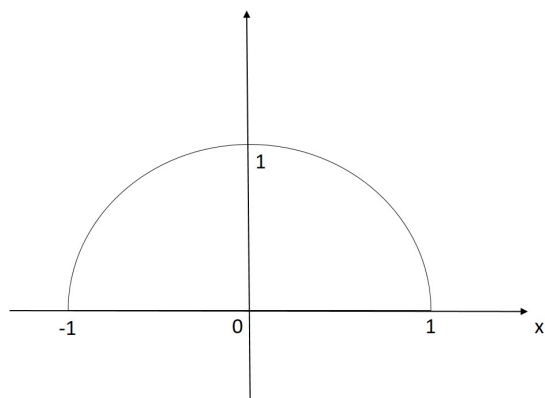


Figure 1: Problem 4 - *semi-circular* Distribution